Product, Quotient, and Trigonometric Derivatives

The product rule

1 Differentiate $f(x) = (3x^2 - 1)(x^3 + 2x)$ using the product rule.

2 Differentiate $f(x) = x^2 \sin x$.

3 Differentiate $f(x) = e^x \cos x$.

The quotient rule

4 Differentiate $f(x) = \frac{3x - 1}{x + 4}$.

5 Differentiate $f(x) = \frac{x^2}{x^2 + 1}$.

6 Differentiate $f(x) = \frac{\sin x}{x}$.

Derivatives of the six trig functions

7 State the derivative of each trig function:

a $\frac{d}{dx}\sin x$ **b** $\frac{d}{dx}\cos x$ **c** $\frac{d}{dx}\tan x$

d $\frac{d}{dx}\cot x$ **e** $\frac{d}{dx}\sec x$ **f** $\frac{d}{dx}\csc x$

8 Differentiate $f(x) = x\tan x$.

9 Differentiate $f(x) = \sec x \cdot \csc x$.

Combining product, quotient, and trig rules

10 Differentiate $f(x) = \frac{e^x}{\sin x}$.

11 Find the equation of the tangent line to $f(x) = x\cos x$ at $x = 0$.

Derivatives using a table of values

12 Given $f(3) = 2$, $f'(3) = -1$, $g(3) = 4$, $g'(3) = 5$, find $h'(3)$ if $h(x) = f(x)g(x)$.

13 Using the same table, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.

Second derivatives with product and quotient rules

14 Find $f''(x)$ for $f(x) = x \sin x$.

More product and quotient rule practice

15 Differentiate $f(x) = (2x + 1)^2(x - 3)$, treating $(2x + 1)^2$ as one factor.

16 Differentiate $f(x) = \frac{x^2 + 1}{x - 1}$.

An applied product-rule problem

17 A company's revenue is $R(t) = p(t) \cdot q(t)$, where $p(t)$ is price and $q(t)$ is quantity sold, both functions of time t (months). At $t = 6$, $p(6) = 20$, $p'(6) = 0.5$, $q(6) = 100$, $q'(6) = -2$. Find $R'(6)$ and interpret its sign.

Higher-order trigonometric derivatives

18 Find $f''(x)$ for $f(x) = \sin x$, and describe the repeating pattern of derivatives f, f', f'', f''', f''''.

19 Using the repeating pattern above, find $f^{(50)}(x)$ for $f(x) = \sin x$ (the 50th derivative).

A capstone combined-rule problem

20 Differentiate $f(x) = \frac{x^2 \sin x}{x + 1}$, applying the quotient rule with the product rule for the numerator's derivative.

21 An object's position is $s(t) = t^2 \cos t$ (meters, seconds). Find the object's velocity function $s'(t)$ using the product rule, then find its velocity at $t = 0$.

22 Differentiate $f(x) = \frac{\cos x}{1 + \sin x}$, and simplify your answer using the Pythagorean identity $\sin^2 x + \cos^2 x = 1$.

23 Given $f(2)=5$, $f'(2)=3$, $g(2)=-1$, $g'(2)=4$, find each of the following at $x=2$:

a $\frac{d}{dx}[f(x)g(x)]$

b $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]$

c $\frac{d}{dx}[3f(x)-2g(x)]$